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School of Electronics Engineering (SENSE)

PROJECT REPORT

COURSE CODE / TITLE	BECE204L– Microprocessors and Microcontrollers		
PROGRAM / YEAR/ SEM	Btech (BEC)/II Year/ WINTER 2023-2024		
DATE OF SUBMISSION	26-04-2024		
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PROJECT TITLE	ULTRASONIC RANGE DETECTOR		
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OBJECTIVE:

The primary objective of this project is to design and implement an ultrasonic range detector using the concepts of the 8051 microcontroller and assembly language programming. The specific goals of the project are as follows:

- *Distance Measurement:* Develop algorithms to accurately measure distances up to a range of 2 meters using an ultrasonic sensor. This involves utilizing the principles of ultrasonic wave propagation time to calculate the distance between the sensor and the obstacle.
- *Hardware Integration:* Integrate the ultrasonic sensor with the 8051 microcontrollers to enable communication and data exchange between the sensor and the controller. This involves configuring the hardware interfaces and pins of the microcontroller to interface with the sensor.
- *Algorithm Development:* Develop algorithms in assembly language to calculate the distance based on the time taken for ultrasonic waves to propagate and reflect back to the sensor. This algorithm should account for factors such as the speed of sound and signal processing overhead.
- *LCD Display:* Implement functionality to display the measured distance on an LCD display. The LCD display will provide a visual representation of the distance measurement, making it easy for the user to interpret the data.
- *Buzzer Alert:* Incorporate a buzzer into the system to provide an audible alert when the measured distance is less than or equal to 10 centimetres. This feature will serve as a safety measure to alert users of close proximity to obstacles.
- *System Reliability:* Ensure the reliability and robustness of the system by conducting thorough testing under various conditions and environments. This includes validating the accuracy of distance measurements, assessing the responsiveness of the LCD display, and evaluating the effectiveness of the buzzer alert feature.

By achieving these objectives, the project aims to demonstrate the capabilities of the 8051 microcontroller and assembly language programming in the development of an ultrasonic range detector system. Additionally, the project seeks to provide a practical solution for accurate distance measurement in various applications such as robotics, automation, and security systems.

INTRODUCTION:

The project focuses on the development of an ultrasonic range detector using the 8051 microcontroller and assembly language programming. Ultrasonic range detectors are widely used in various applications such as robotics, automotive, and industrial automation for accurate distance measurement. The existing ultrasonic range detectors typically utilize microcontrollers and programming languages like C or Arduino IDE for implementation.

Existing:

The existing ultrasonic range detectors rely on microcontrollers and high-level programming languages such as C or Arduino IDE. These systems are effective but may have limitations in terms of flexibility, customization, and resource utilization. Additionally, they may lack features such as real-time processing, low-level hardware control, and efficient memory usage.

Motivation:

The motivation behind undertaking this project stems from the desire to explore the capabilities of the 8051 microcontroller and assembly language programming in the development of embedded systems. The 8051 microcontroller is widely used in embedded applications and offers advantages such as low cost, low power consumption, and ease of integration. By delving into assembly language programming, we aim to gain a deeper understanding of system-level programming and hardware interfacing.

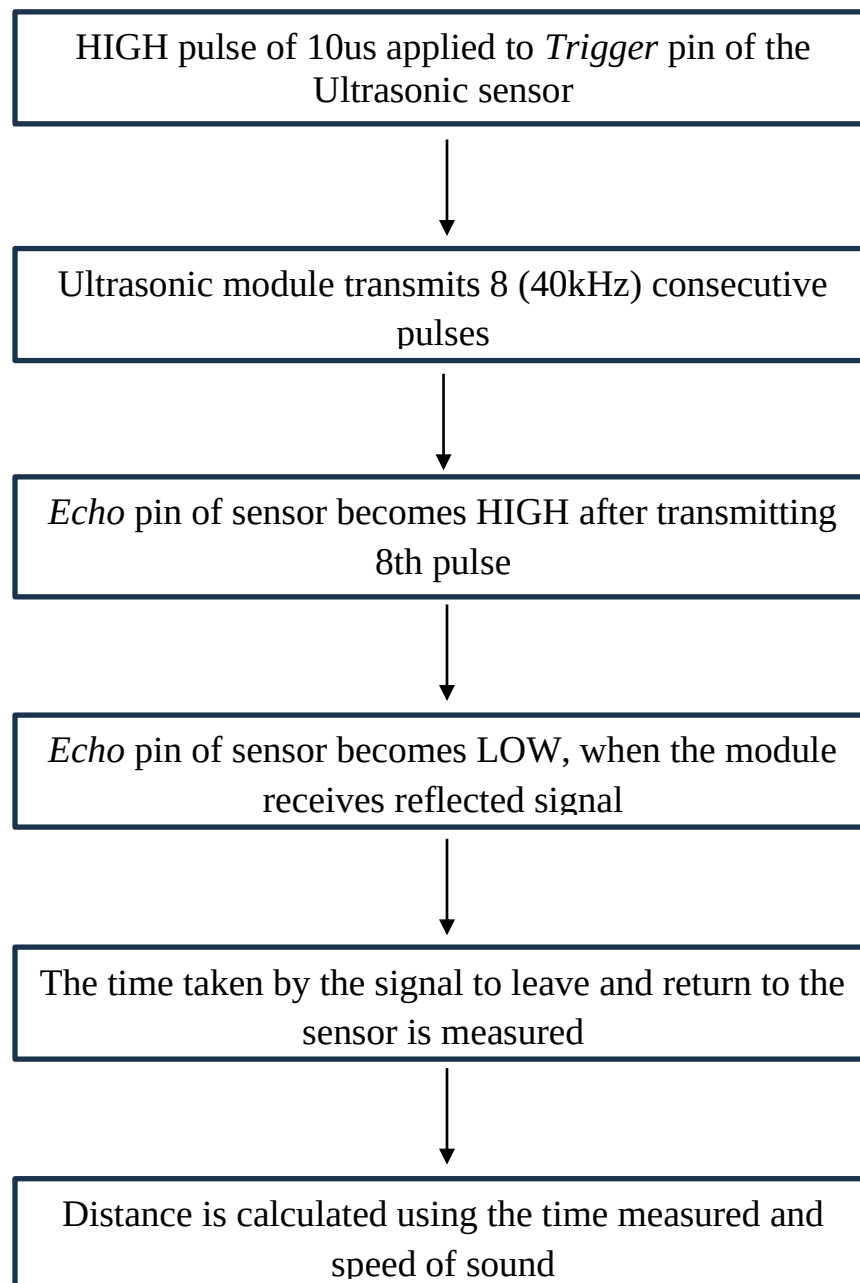
Difference from Existing:

Our work differs from existing ultrasonic range detectors in several aspects:

1. **Utilization of the 8051 Microcontroller:** We utilize the 8051 microcontroller, which offers unique features and capabilities compared to other microcontrollers. This allows for a different approach to system design and implementation.
2. **Assembly Language Programming:** Unlike existing systems that often use high-level programming languages, we employ assembly language programming for low-level hardware control and optimization. This enables us to achieve greater efficiency and performance in our system.
3. **Customization and Flexibility:** Our system provides greater customization and flexibility due to the direct control over hardware resources afforded by assembly language programming. This allows for tailored solutions to specific application requirements.

4. Real-time Processing: With assembly language programming, we can implement real-time processing algorithms for time-sensitive applications such as obstacle detection and avoidance.
5. Overall, our work aims to showcase the capabilities of the 8051 microcontroller and assembly language programming in the development of ultrasonic range detectors, offering a unique perspective and potential advantages over existing systems.

BLOCK DIAGRAM:



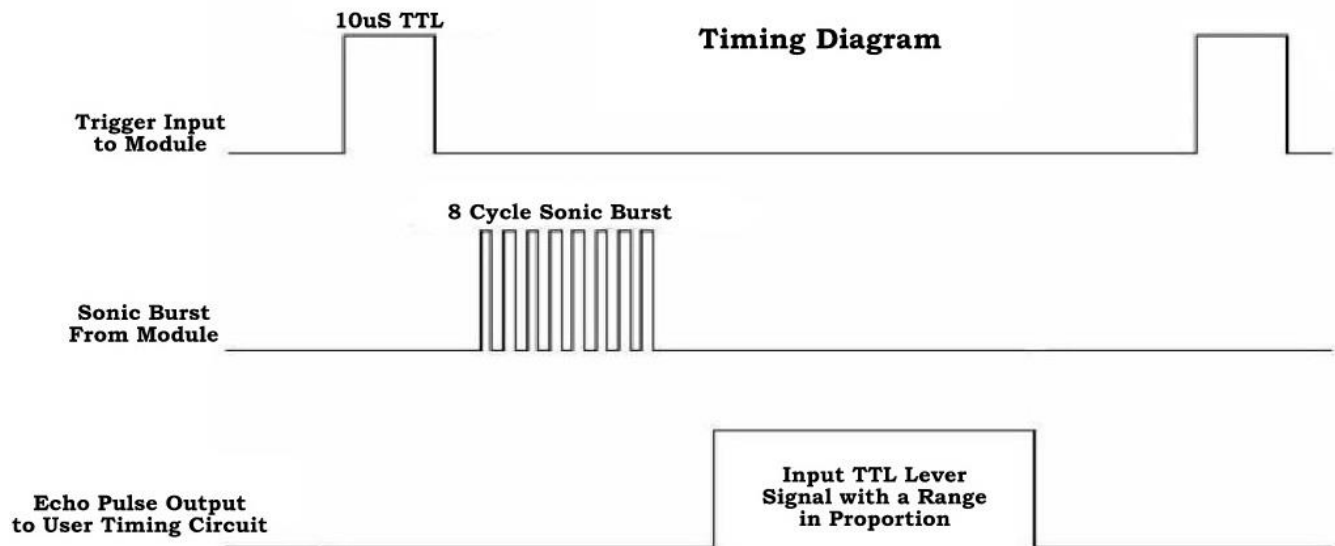
COMPONENTS/ SOFTWARE REQUIRED:

Components/Software	Application
8051 Microcontroller Kit	Acts as the central processing unit for the ultrasonic range detector system, facilitating control and communication with other components.
HC-SR04 Ultrasonic Module	Provides the ability to transmit and receive ultrasonic waves for distance measurement.
16x2 LCD Display	Displays the measured distance in a user-readable format for easy interpretation.
10K Ω Potentiometer	Used for adjusting the contrast and brightness of the LCD display for optimal visibility.
Push Button	Enables user input for initiating specific actions or functions within the system.
Breadboard	Provides a convenient platform for prototyping and connecting electronic components during development.
Buzzer	Produces audible alerts to notify users when the measured distance is less than or equal to 10 centimetres.
Connecting Wires	Used for establishing electrical connections between various components in the circuit.
Power Supply	Supplies the necessary voltage and current to power the entire system and its components.
Proteus Software	Utilized for simulating and testing the functionality of the circuit design before hardware implementation.
Keil μ Vision Software	Used for writing, compiling, and debugging the assembly language code for the 8051 microcontroller.
Crystal Oscillator	A crystal oscillator may be required to provide clock signals to the microcontroller for precise timing and synchronization. In our project, the crystal oscillator was pre-built into the 8051 microcontroller kit.

PROJECT DESCRIPTION:

The distance measurement in this project utilizes the pulse echo method. Here, the ultrasonic module emits a signal towards the object, which then reflects the signal back to the module. The module detects this echo signal and produces an output signal, the time period of which is directly proportional to the distance of the object. This mechanism is akin to RADAR (Radio Detection and Ranging) systems, where distance is determined based on the time taken for signals to travel to and from objects.

The circuit operates by transmitting a 10 μ s HIGH pulse to the TRIG pin of the ultrasonic module. This triggers the module to emit 8 consecutive pulses at a frequency of 40 KHz. Upon transmitting the 8th pulse, the ECHO pin of the sensor becomes HIGH. Subsequently, when the module receives the reflected signal from the object, the ECHO pin transitions to LOW. The time taken for the signal to travel from the module, reflect off the object, and return to the module is utilized to calculate the range of the object.



The distance to the object is determined using the formula: Distance (in centimetres) = Time taken (in microseconds) / 58, or alternatively, Distance (in inches) = Time taken (in microseconds) / 148. Additionally, the distance can also be calculated using the speed of the ultrasonic wave, which is typically around 340 meters per second.

The calculated distance is then displayed on the LCD display in a user-readable format. The LCD display provides real-time feedback to the user about the distance between the sensor and the object, enhancing situational awareness and facilitating navigation in various applications.

Additionally, if the measured distance falls below a predefined threshold, typically 10 centimetres, the microcontroller activates the buzzer to emit an audible alert. This alert notifies the user of close proximity to obstacles, thereby enhancing safety and preventing collisions or accidents.

The system operates continuously in a loop, continuously triggering the ultrasonic sensor, measuring distances, updating the LCD display, and monitoring for proximity alerts. This ensures that the user receives real-time feedback about the surroundings and potential obstacles, facilitating safe and efficient operation in various environments.

In conclusion, the ultrasonic range detector project effectively measures distances, provides real-time feedback to the user, and enhances safety in various applications. Through its robust design and functionality, it serves as a valuable tool for navigation, obstacle detection, and proximity monitoring in diverse scenarios.

SIMULATION:

In Proteus, the design of the ultrasonic range detector project involves several steps, including component selection, schematic design, simulation setup, and result analysis. Let's break down each step in detail:

1. Component Selection:

- Open Proteus software and create a new project.
- Navigate to the Components mode and select the required components for the project, including:
 - 8051 Microcontroller
 - HC-SR04 Ultrasonic Sensor
 - 16x2 LCD Display
 - Buzzer
 - Potentiometer (for LCD contrast adjustment)
 - Push Button (for triggering distance measurement)
 - Power Supply
 - Connecting wires

2. Schematic Design:

- Once the components are selected, switch to the Schematic Capture mode.
- Place the components onto the schematic canvas and wire them together according to the project requirements.
- Connect the following pins of the components:
- Connect TRIG pin of the ultrasonic sensor to P3.1 pin of the microcontroller.
- Connect ECHO pin of the ultrasonic sensor to P3.0 pin of the microcontroller.
- Connect the data pins of the LCD display to Port 2 of the microcontroller.
- Connect the buzzer to P3.2 pin of the microcontroller.
- Connect the RS, RW and E pins of the LCD to P3.7, P3.6 and P3.5 pins respectively.
- Ensure proper power supply connections to all components.

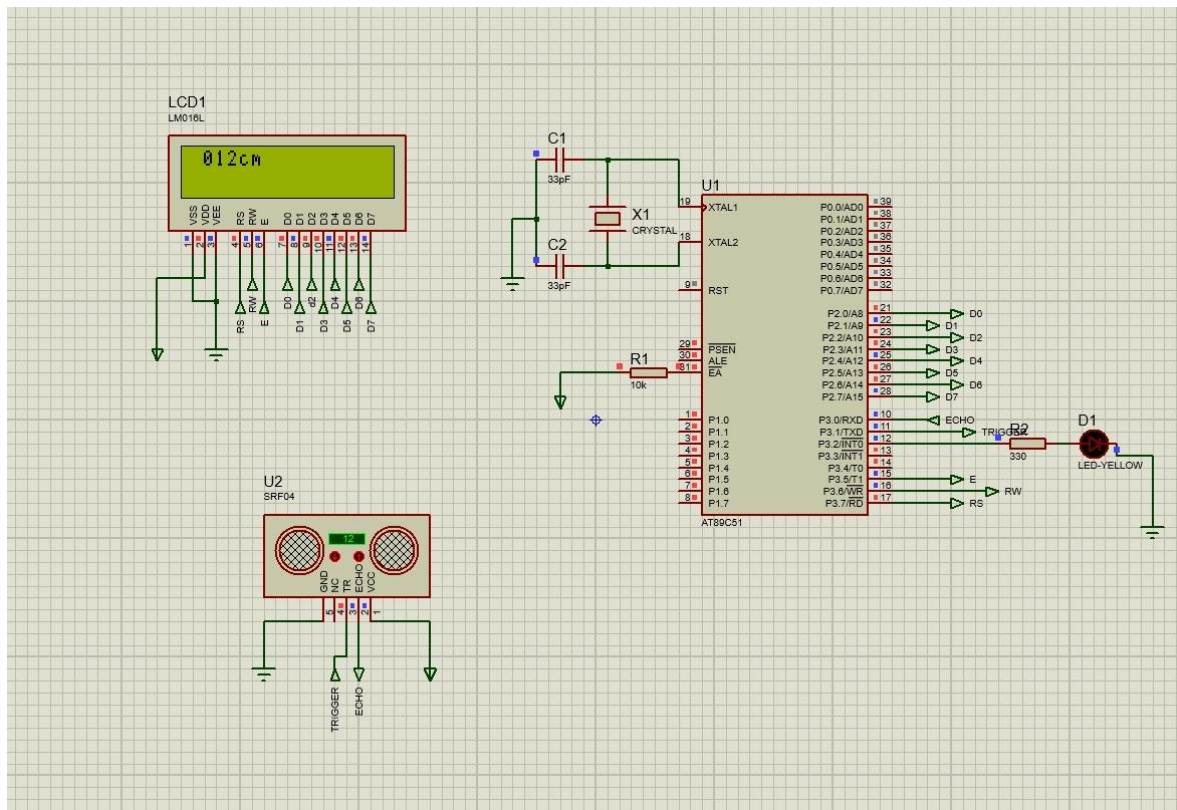
3. Simulation Setup:

- After designing the schematic, switch to the Simulation mode.
- Configure the microcontroller by double-clicking on it and setting its properties, such as clock frequency and program file.
- Write the firmware code for the microcontroller to control the ultrasonic sensor, LCD display, and buzzer.
- Add appropriate simulation models for the ultrasonic sensor, LCD display, and buzzer from the Proteus library.
- Simulate the interaction between the microcontroller and the components to ensure proper functionality.

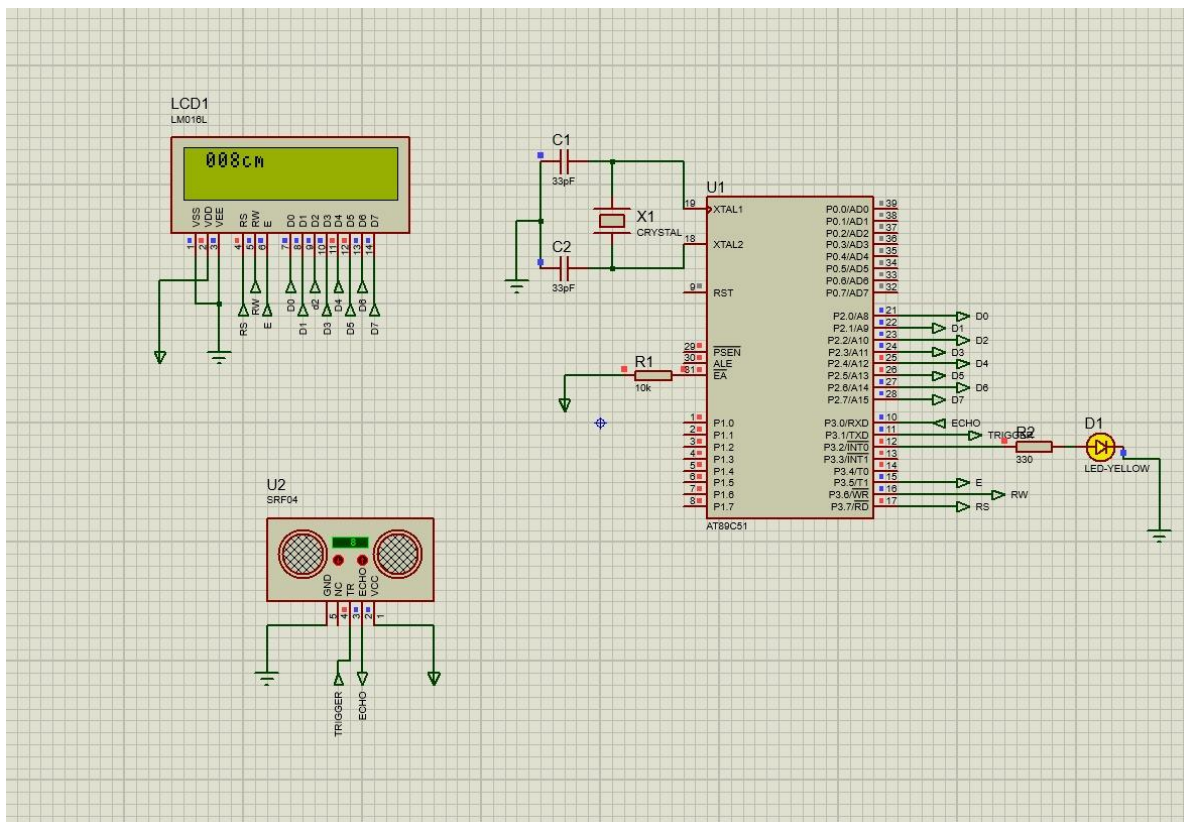
4. Result Analysis:

- Run the simulation to observe the behaviour of the ultrasonic range detector system.
- Verify that the microcontroller triggers the ultrasonic sensor correctly, measures distances accurately, and displays the results on the LCD display.
- Check if the buzzer activates when the measured distance is less than or equal to the predefined threshold.

Case1: Distance greater than 10cm



Case2: Distance lesser than or equal to 10cm



CONCEPT LEARNED:

In this project we learnt about the fundamental concepts of ultrasonic range detection through the integration of ultrasonic sensors with the 8051-microcontroller. It explores the principles of ultrasonic wave propagation, signal processing, and distance measurement techniques. By implementing these concepts on the 8051-microcontroller platform, it provides a comprehensive understanding of ultrasonic sensing technology and its applications.

We learnt about the principles of ultrasonic distance measurement, including the pulse echo method used by ultrasonic sensors to calculate distances based on the time taken for ultrasonic waves to travel to and from objects. This project taught me how to interface a microcontroller, specifically the 8051, with various peripheral devices such as an ultrasonic sensor, LCD display, and buzzer. I learned how to control these devices using GPIO pins and write firmware code to manage their operation.

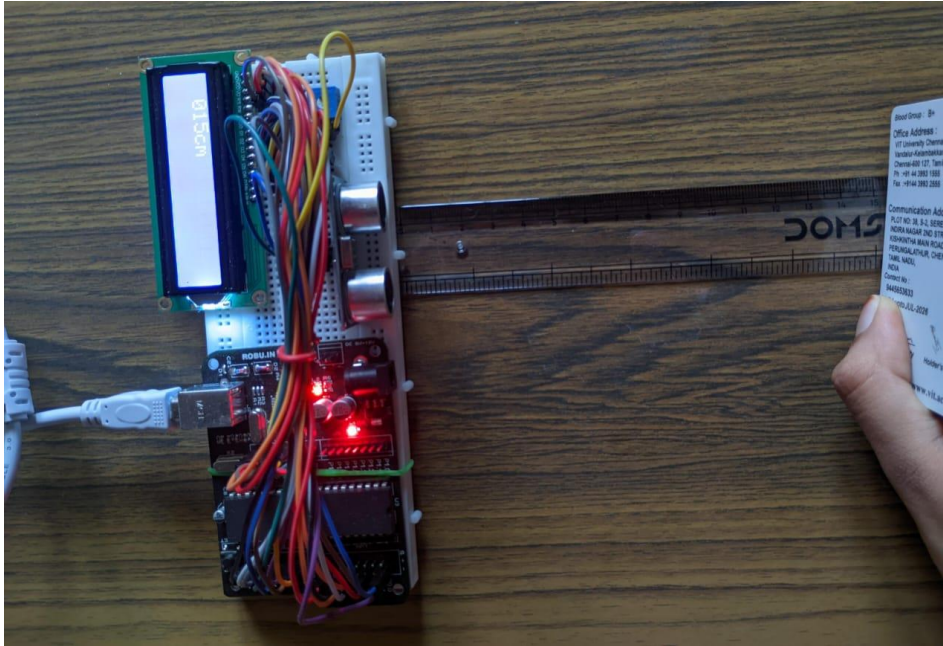
I gained experience in integrating sensors into electronic systems and understanding their data output formats. In this project, I learned how to interface the HC-SR04 ultrasonic sensor and interpret its echo signal to measure distances accurately. The project emphasized the importance of real-time feedback in electronic systems, especially for applications like distance measurement. I learned how to display real-time data on an LCD display and trigger alerts using a buzzer based on measured distances.

Through simulation in Proteus, I learned how to test and validate electronic designs before hardware implementation. I gained experience in setting up simulations, writing firmware code, and analysing simulation results to ensure proper functionality of the project.

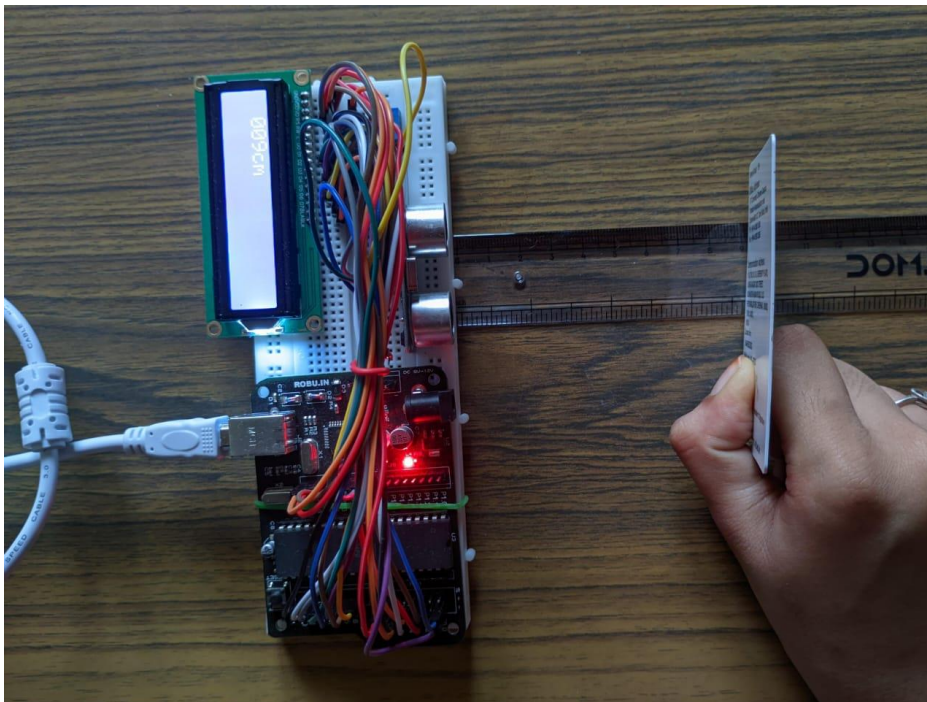
Throughout the project, I encountered various challenges and learned valuable troubleshooting and debugging techniques. I gained experience in identifying and resolving issues such as incorrect sensor readings, display anomalies, and firmware errors, which enhanced my problem-solving skills. This project helped me understand the intricate interaction between hardware and software components in embedded systems. I learned how firmware code controls hardware peripherals and how hardware constraints influence software design decisions, leading to a deeper appreciation of system-level design considerations.

IMPLEMENTATION:

Case1: Distance greater than 10cm



Case2: Distance lesser than or equal to 10cm



The buzzer would turn on when the distance is less than 10 cm.

YouTube video link: <https://www.youtube.com/watch?v=GV6LeQeAool>

CHALLENGES FACED:

One of the primary challenges encountered in this project is hardware interfacing. Interfacing with external hardware components such as the ultrasonic sensor and the LCD display requires meticulous attention to detail. Ensuring proper electrical connections, voltage levels, and signal timings can be challenging, particularly when dealing with different hardware modules and interfaces. The challenge was calibrating the ultrasonic sensor to ensure accurate distance measurements. It required fine-tuning the timing parameters and understanding the sensor's response to different environmental conditions.

The complexity of assembly language programming presents a unique challenge when working with the 8051 microcontrollers. Writing efficient and error-free code requires a deep understanding of the microcontroller's architecture and instruction set. Debugging code running on the 8051 microcontroller can also be challenging due to limited debugging capabilities, necessitating thorough testing and validation.

Timing and synchronization present another critical challenge. The timing requirements for triggering the ultrasonic sensor, capturing the echo pulse, and updating the LCD display need to be precisely synchronized to ensure accurate distance measurements and reliable display output. Achieving this synchronization without any timing conflicts or glitches can be particularly challenging, given the limited resources and capabilities of the 8051 microcontrollers.

In conclusion, integrating an ultrasonic sensor and an LCD display with an 8051 microcontroller presents several significant challenges, including hardware interfacing, timing and synchronization, resource constraints, calibration and testing, assembly language programming complexity, and signal processing challenges. Addressing these challenges requires careful planning, thorough testing, and iterative refinement to ensure the system operates reliably and accurately in real-world applications.

APPLICATIONS:

The ultrasonic range detector project offers versatility and utility across various applications due to its ability to accurately measure distances and provide real-time feedback. Some of the applications where this project can be useful include:

1. *Robotics*: In robotics, the ultrasonic range detector can serve as a critical component for obstacle detection and avoidance. Robots equipped with this technology can navigate complex environments autonomously, ensuring safe operation and avoiding collisions with obstacles.
2. *Home Automation*: In home automation systems, the ultrasonic range detector can be used to monitor room occupancy, detect intruders, or trigger automated actions based on the presence or absence of individuals. For example, it can turn on lights when someone enters a room and turn them off when the room is vacant.
3. *Security Systems*: The project can be integrated into security systems to detect unauthorized entry or intrusion. By placing ultrasonic sensors at key entry points, such as doors and windows, the system can detect movement and trigger alarms to alert occupants or security personnel.
4. *Industrial Automation*: In industrial settings, the project can be utilized for proximity sensing, object detection, and automated material handling. It can help optimize manufacturing processes, improve safety, and prevent accidents by detecting the presence of objects or individuals in hazardous areas.
5. *Parking Assistance*: In automotive applications, the ultrasonic range detector can be employed for parking assistance systems to aid drivers in maneuvering and parking their vehicles safely. By detecting obstacles and providing distance feedback, the system assists drivers in avoiding collisions and parking accurately.
6. *Smart Agriculture*: In agriculture, the project can be integrated into precision farming systems for crop monitoring and irrigation management. By measuring distances between plants or monitoring soil moisture levels, the system helps optimize crop yields and conserve water resources.

Overall, the ultrasonic range detector project offers immense potential for enhancing safety, efficiency, and automation across various domains, making it a valuable tool for diverse applications.

CONCLUSION:

In conclusion, the ultrasonic range detector project successfully demonstrates the implementation of a versatile and valuable electronic system capable of accurately measuring distances and providing real-time feedback. By utilizing the pulse echo method and interfacing with a microcontroller, the project achieves precise distance measurements using ultrasonic waves. The integration of components such as the ultrasonic sensor, LCD display, and buzzer enables the system to offer functionality beyond basic distance measurement, including obstacle detection and alerting. Through simulation and testing in Proteus, the project ensures reliability, efficiency, and effectiveness before hardware implementation. With applications spanning robotics, home automation, security systems, industrial automation, healthcare, agriculture, retail analytics, and more, the project showcases the versatility and utility of ultrasonic range detectors in various domains. Overall, the project underscores the importance of innovation, problem-solving, and interdisciplinary collaboration in engineering and technology fields, contributing to the advancement of electronic systems and their applications in diverse industries.

However, it's important to remember that ultrasonic sensors have limitations. They can be affected by temperature fluctuations, air movement, and the reflective properties of objects. For high-precision applications, other sensor technologies might be more suitable.

Despite these limitations, ultrasonic range detectors with 8051 microcontrollers remain a valuable tool for many projects due to their affordability, ease of use, and wide range of potential applications.

APPENDICES:

APPENDIX-A: Code

TRIG EQU P3.1	Labelling P3.1 as TRIG
ECHO EQU P3.0	Labelling P3.0 as ECHO
E EQU P3.5	labelling P3.5 as enable
RS EQU P3.7	Labelling P3.7 as RS
RW EQU P3.6	Labelling P3.6 as RW
LCD_data EQU P2	Labelling P2 as LCD_data
LED EQU P3.2	Labelling P3.2 as LED
ORG 0000H	Origin
SETB ECHO	Setting as input pin
CLR TRIG	Setting as output pin
MOV TMOD, #02H	Timer 0 - Mode 2
MOV TH0, #198D	
ACALL LCD_initialization	Calling LCD_initialization subroutine
Loop1: ACALL Get_data	Calling Get_data subroutine
ACALL Buzz	Calling Buzz subroutine
ACALL Convert	Calling Convert subroutine
ACALL Cursor_home	Calling Cursor_home subroutine
ACALL Display	Calling Display subroutine
SJMP Loop1	Running the loop infinitely
LCD_initialization:	
MOV DPTR, #Command	Moving command data to DPTR
CLR RS	Clearing RS for command transmission
CLR RW	Clearing RW for command transmission
L1: CLR A	Clear Accumulator
MOVC A,@A+DPTR	Move DPTR data to Accumulator
JZ Return	Jump on Zero to Return
SETB E	High to low pulse for enable pin
MOV LCD_data, A	Moving Accumulator data to P1
CLR E	
ACALL delay1ms	Calling delay subroutine
INC DPTR	Incrementing DPTR value
SJMP L1	Repeating the loop
Command: DB 38H,0EH,01H,06H,80H,0	Command Data bit
Return: RET	Return from subroutine
Cursor_home:	
CLR RS	Clearing RS
SETB E	High to Low pulse for Enable
MOV LCD_data,#80H	Row 1 - 1st position
CLR E	Clearing enable
ACALL delay10ms	Calling delay subroutine

SETB E	High to Low pulse for Enable
MOV LCD_data,#0CH	Display ON, Cursor OFF
CLR E	Clearing enable
RET	Return from subroutine
Get_data:	
CLR A	Clearing Accumulator
SETB TRIG	Sending a high pulse of 10us to trigger pin
ACALL delay_10us	Calling delay subroutine
CLR TRIG	Clearing trigger pin
Here: JNB ECHO, Here	Wait for echo to become high
SETB TR0	Start the timer
Here2: JNB TF0, Here2	Wait for timer flag to become high
INC A	Increments A for every timer0 overflow
CLR TF0	Clear timer flag
JZ Return2	Jump on zero to return
JB ECHO, Here2	Jumps to Here2 if echo is still available
CLR TR0	Stop the timer
Return2:RET	Return from subroutine
delay_10us:	10us Delay subroutine
MOV R7, #18	
Stay: DJNZ R7, Stay	
RET	
Buzz: CJNE A,#10D,Label1	Comparing Accumulator value with 10
CLR LED	Setting LED if A=10
SJMP EXIT	Exit to return
Label1: JNC Label2	Checking carry
CLR LED	Clearing LED if A>10
SJMP EXIT	Exit to return
Label2: SETB LED	Setting LED if A<10
EXIT: RET	Return from subroutine
Convert:	
MOV B,#10D	Moving value 10 to B register
DIV AB	Divide A with B
MOV 41H,B	Save low (ONES) digit in 41 RAM ADDRESS
MOV B,#10D	
DIV AB	
MOV 42H,B	Save tenth place digit in 42 RAM ADDRESS
MOV 43H,A	Save Hundredth place digit in 43 RAM ADDRESS
ACALL LOOKUP	Calling LoopUp subroutine
MOV 43H,A	Save the character in 43 RAM location
MOV A,42H	Move data in 42 to A
ACALL LOOKUP	Calling LoopUp subroutine
MOV 42H,A	Save the character in 42 RAM location
MOV A,41H	Move data in 41 to A
ACALL LOOKUP	Calling LoopUp subroutine

MOV 41H,A	Save the character in 41 RAM location
RET	Return from subroutine
LOOKUP:	
CJNE A,#00H,ONE	Checking if the number is 0
MOV A,#'0'	
RET	
ONE: CJNE A,#01H,TWO	Checking if the number is 1
MOV A,#'1'	
RET	
TWO: CJNE A,#02H,THREE	Checking if the number is 2
MOV A,#'2'	
RET	
THREE: CJNE A,#03H,FOUR	Checking if the number is 3
MOV A,#'3'	
RET	
FOUR: CJNE A,#04H,FIVE	Checking if the number is 4
MOV A,#'4'	
RET	
FIVE: CJNE A,#05H,SIX	Checking if the number is 5
MOV A,#'5'	
RET	
SIX: CJNE A,#06H,SEVEN	Checking if the number is 6
MOV A,#'6'	
RET	
SEVEN: CJNE A,#07H,EIGHT	Checking if the number is 7
MOV A,#'7'	
RET	
EIGHT: CJNE A,#08H,NINE	Checking if the number is 8
MOV A,#'8'	
RET	
NINE: CJNE A,#09H,TEN	Checking if the number is 9
MOV A,#'9'	
RET	
TEN: RET	
Display:	
CLR RW	Clearing RW for data transmission
SETB RS	Setting RS for data transmission
ACALL delay1ms	Calling delay subroutine
SETB E	High to low pulse for enable pin
MOV LCD_data,#' '	Moving blank space to LCD
CLR E	Clearing enable
ACALL delay1ms	Calling delay subroutine
SETB E	High to low pulse for enable pin
MOV LCD_data,43H	Moving data in 43 to LCD
CLR E	Clearing enable

ACALL delay1ms	Calling delay subroutine
SETB E	High to low pulse for enable pin
MOV LCD_data,42H	Moving data in 42 to LCD
CLR E	Clearing enable
ACALL delay1ms	Calling delay subroutine
SETB E	High to low pulse for enable pin
MOV LCD_data,41H	Moving data in 41 to LCD
CLR E	Clearing enable
ACALL delay1ms	Calling delay subroutine
SETB E	High to low pulse for enable pin
MOV LCD_data,'#c'	Moving character 'c' to LCD
CLR E	Clearing enable
ACALL delay1ms	Calling delay subroutine
SETB E	High to low pulse for enable pin
MOV LCD_data,'#m'	Moving character 'm' to LCD
CLR E	Clearing enable
ACALL delay1ms	Calling delay subroutine
RET	return from subroutine
delay10ms: MOV R3,#1	Delay subroutine for 10ms
MOV R2,#1	
MOV R1,#19	
TT1: DJNZ R1,TT1	
DJNZ R2,TT1	
DJNZ R3,TT1	
RET	
delay1ms: MOV R2,#04	Delay subroutine for 1ms
MOV R1,#18	
TT2: DJNZ R1,TT2	
DJNZ R2,TT2	
RET	
delay_2s: MOV R3,#50	Delay subroutine for 2s
MOV R2,#10	
MOV R1,#250	
TT3: DJNZ R1,TT1	
DJNZ R2,TT1	
DJNZ R3,TT3	
RET	
END	THE END

APPENDIX-B: Hardware Implementation

